Matching Advice on Degree-Two Bipartite Graphs: Exact Consistency–Robustness Frontiers

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We study unweighted online bipartite matching when every realized vertex has degree at most two and advice is an injective partial matching table that may contain absent edges. Against an oblivious adversary, we determine the largest robustness attainable at every requested consistency c: it is $\min\{3/4, 3/2 - c\}$ on this class and $\min\{7/8, 7/4 - c\}$ on its 2-regular subclass. Consistency holds for every maximum matching supplied as advice. The central algorithm gives priority to available offline neighbors whose second incident edge has appeared, and uses advice only among fresh neighbors. It is optimal under maximum-matching advice and retains a deterministic $3/4$ guarantee on cycles under arbitrary advice. Combining this endpoint with the known $7/8$ cycle baseline gives the cycle frontier; common advice tables on small indistinguishable inputs give matching upper bounds.

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1 Introduction

A proposed matching can guide an online allocation rule, but the proposal may be wrong about edges that have not yet appeared. Robustness measures performance for every such proposal; consistency measures performance when the proposal is an optimum matching. We determine their exact tradeoff for unweighted bipartite graphs of maximum degree two on both sides. This restriction separates paths from cycles and makes the cost of using advice explicit: perfect consistency permits robustness $1/2$ when paths are allowed, and $3/4$ when every component is a cycle.

The structural reason for the stronger cycle guarantee is local. Once an offline vertex has appeared twice, it cannot be used by a future arrival. Prioritizing that vertex over a fresh one protects future options. We show that advice can break the remaining ties without sacrificing optimality under *any* maximum advised matching, even when it is partial. On cycles, every early choice creates one occupied and one free boundary. A later failure consumes two occupied boundaries, so arbitrary advice still leaves a deterministic $3/4$ matching. This advice-preserving property is the main step in the tradeoff characterization.

Model and guarantees. Let U, V be labelled sets of size n, with U offline and V online. The realized graph $G = (U, V, E)$ is simple and unweighted. Each $v \in V$ arrives once in an arbitrary order π, revealing its whole neighborhood. An online algorithm irrevocably matches v to an unused neighbor, or leaves it unmatched. Its output is an integral matching. The algorithm knows U, n, and the online identifiers used in the advice, but does not know future neighborhoods, their order, or the component decomposition.

The complete advice is a table $A : V \to U \cup \{\perp\}$ with distinct non-$\perp$ values. It specifies a partial matching in the complete bipartite graph on the given labels. An advised pair need not be an edge of G. There are no annotations or additional graph information in the advice. The graph, order, and table are fixed independently of the algorithm's private randomness: the adversary is oblivious. Write $\mathrm{OPT}(G)$ for maximum matching cardinality. An algorithm is c-consistent if, for every G, π and *every* table A that is a maximum matching of G,

$$\mathbb{E}|\mathsf{ALG}(G, \pi, A)| \geq c\,\mathrm{OPT}(G).$$

It is r-robust if the same inequality with r holds for every allowed table. All guarantees are uniform over finite n and have no additive loss; inputs with $\mathrm{OPT}(G) = 0$ impose no condition.

Let $\mathcal{D}$ contain the graphs of maximum degree at most two on both sides, including isolated vertices. Let $\mathcal{C} \subset \mathcal{D}$ contain the 2-regular graphs, which are disjoint unions of simple even cycles, including four-cycles. For a class $\mathcal{X}$, let $r_{\mathcal{X}}(c)$ be the largest robustness attainable by a c-consistent randomized online algorithm on that class.

Theorem 1. *For every $c \in [0, 1]$,*

$$r_{\mathcal{D}}(c) = \min\{3/4, 3/2 - c\}, \qquad r_{\mathcal{C}}(c) = \min\{7/8, 7/4 - c\}.$$

Both values are attained in the model above.

Closest results and the contribution. Fàbrega and Muñoz [FM18], Propositions 1–2, already establish the advice-free $7/8$ optimum on cycles, the eight-cycle obstruction used below, and the edge-orientation argument underlying the cycle analysis. Their ratio uses the reciprocal convention. Feng, Li, Wu, and Zhou [FLWZ25], Algorithm 2 and its subsequent discussion, identify the relevant priority rule as a degree-two semi-OCS rule. We credit those baselines and the orientation method. Our proof includes them to make the path extension and the effect of advice self-contained.

Choo, Gouleakis, Ling, and Bhattacharyya [CGLB24], Section 3, already give the Z-path obstruction and the linear bound $c + r \leq 3/2$ in the broader adversarial-arrival setting. Jin and Ma [JM24], Proposition 3.1, also use this obstruction and obtain the same numerical unweighted frontier in a two-stage model. Their algorithm sees arrivals in batches and their consistency benchmark uses the value of following first-stage advice and completing later. Here every online choice is irrevocable before the next neighborhood appears, and consistency concerns every optimum full matching table. The contribution on $\mathcal{D}$ is the sequential

achievability argument; on C it is the advice-preserving endpoint and the resulting exact frontier. Convex mixing itself is standard.

Two related distinctions matter. Bhangale, Chakraborty, and Harsha [BCH25] bound only the *online* degree in their degree-two integral theorem; their separate 3/4 theorem allows fractional actions. Lindermayr, Megow, and Paul [LMP26] study multistage allocation and also give a one-by-one tradeoff for vertex-weighted *fractional* matching. An integral prediction in that model does not make its actions integral. These statements therefore address different algorithmic classes from Theorem 1.

2 Priority rules and the advice-preserving endpoint

Maintain $h(u)$, the number of revealed incident edges at $u \in U$. On arrival of v, increment $h(u)$ for every $u \in N(v)$. Call a neighbor *fresh* if $h(u) = 1$, and *exhausted* if $h(u) = 2$. An exhausted neighbor has no future incident edge. A neighbor is available when it is unmatched. Fix a label order for deterministic ties.

Both rules below first take an available exhausted neighbor, if one exists, and leave v unmatched only when no neighbor is available.

- E ignores the table. Without an available exhausted neighbor, it takes the sole available neighbor, or chooses fairly between two fresh neighbors, independently at each such choice.
- T uses the same priority. Without an available exhausted neighbor, it takes $A(v)$ if that is an actual available neighbor; otherwise it takes the first available neighbor in the fixed order.

These rules use only the revealed prefix and the matching table. In particular, an absent advised edge is simply ineligible.

Lemma 2. *For every maximum matching table A of a graph in $\mathcal{D}$, the output of* T *has cardinality* $\mathrm{OPT}(G)$.

Proof. Let M be the output and suppose $|M| < |A|$. The symmetric difference of A and M contains an M-augmenting path

$$v_0, u_0, v_1, u_1, \ldots, v_k, u_k,$$

where $v_i u_i \in A$, $v_{i+1} u_i \in M$, and v_0, u_k are unmatched in M. If $k = 0$, greediness gives an immediate contradiction. Otherwise u_0 was already occupied when v_0 arrived, since v_0 remained unmatched. Thus its partner v_1 arrived earlier than v_0.

Inductively suppose v_i arrived before v_{i-1}, for $i \geq 1$. The two neighbors of u_{i-1} are precisely v_i and v_{i-1}, by the offline degree bound, so u_{i-1} was fresh when v_i selected it. Consequently no exhausted neighbor was available then. If its advised neighbor u_i had been free, T would have preferred u_i to u_{i-1}: either by exhausted priority or by the advice tie-break. Thus u_i was already occupied. For $i < k$ its partner v_{i+1} arrived earlier still. At $i = k$ this is impossible, since u_k is free even at termination. The resulting strictly decreasing arrival times contradict the existence of the augmenting path. □

The argument applies to every maximum table, including partial ones. The offline degree bound, rather than perfection or uniqueness of the advice, is what makes the selected neighbor fresh along the augmenting path.

3 Boundary accounting on paths and cycles

We analyze final components only for the proof; the algorithm never needs to identify them. Their arrivals can be arbitrarily interleaved.

Within a nontrivial component, classify a degree-two arrival as a *root*, *extension*, or *merge* according as it has zero, one, or two previously seen neighbors. Let their counts be a, b, z. A degree-one online vertex is an *initial leaf* if its neighbor is fresh, and a *terminal leaf* otherwise; let their counts be s, t. These events are determined by the fixed graph and order.

An offline vertex seen once is a boundary of the revealed component. Mark it with bit 1 if occupied and bit 0 if free. A root creates two complementary boundary bits. Under E they are assigned by that root's independent fair coin. An initial leaf creates one deterministic bit 1. At an extension the old neighbor becomes exhausted. If free, it is chosen and the new boundary stays free; if occupied, the new neighbor is

chosen and the new boundary is occupied. Thus the extension always succeeds and transports the old bit unchanged.

A merge consumes its two boundaries and fails exactly when both bits are 1. Its choice between free exhausted neighbors cannot affect a future boundary. A terminal leaf consumes one bit and fails exactly when it is 1. Every surviving boundary is therefore an original root or initial-leaf bit transported without copying. Its source is fixed by the graph and order, independently of all matching outcomes.

Contract extensions. For a final path, the resulting graph of roots and initial leaves linked by merges is a path, with terminal leaves or final offline endpoints at any remaining ports. For a final cycle it is a cycle of roots. In particular, a merge between distinct root objects under E fails with probability $1/4$, and a merge between an initial leaf and a root fails with probability $1/2$. These are unconditional probabilities on a fixed input. No independence conditional on earlier successes is needed.

Lemma 3. *Let a nontrivial path have m online vertices and $\ell \in \{0, 1, 2\}$ online endpoints. If $a \geq 1$, then*

$$\mathbb{E}|\mathsf{E}(path)| = m - \frac{a-1}{4} - \frac{\ell}{2}. \tag{1}$$

If $a = 0$, E is optimal on the path. On a cycle with m online vertices,

$$\mathbb{E}|\mathsf{E}(cycle)| = \begin{cases} m, & a = 1, \\ m - a/4, & a \geq 2. \end{cases} \tag{2}$$

Proof. For a path with q offline vertices, count first appearances and all arrivals to obtain

$$q = m + 1 - \ell = 2a + b + s, \quad s + t = \ell, \quad m = a + b + z + s + t, \quad z = a + s - 1. \tag{3}$$

Suppose $a \geq 1$. In the contracted path, an initial-leaf object has only one port. Joining two such objects would complete an entire component without a root, so each of the s initial leaves is joined to a root. The remaining $z - s = a - 1$ merge edges join distinct roots. Likewise a terminal leaf consumes a root bit: connecting it directly, after extensions, to an initial leaf would complete a separate rootless path. The expected failures are therefore $(a-1)/4 + s/2 + t/2$, giving (1).

If $a = 0$, the first arrival is an initial leaf and every boundary is occupied. Roots and extensions cannot fail; all z merges and t terminal leaves fail. There are $z + t = \ell - 1$ failures, giving $m - \ell + 1 = \min\{m, q\}$ matches, the path optimum. Isolated vertices contribute zero and are handled separately.

For a cycle, $s = t = 0$, $m = q = 2a + b$, and $z = a$. If $a = 1$, the closing merge sees the complementary bits of its sole root and succeeds. If $a \geq 2$, every edge of the contracted root cycle joins distinct roots: a self-join would already close a separate final cycle. Every merge then fails with probability $1/4$. With two roots the contracted cycle has two parallel edges, each using different ports. Linearity of expectation still gives (2), without requiring independence between merge failures. □

Corollary 4. *The rule* E *guarantees* $3\,\mathrm{OPT}/4$ *on* $\mathcal{D}$ *and* $7\,\mathrm{OPT}/8$ *on* C.

Proof. Rootless paths are optimal. For a path with $a \geq 1$ and $\ell = 0$, $\mathrm{OPT} = m = 2a + b - 1$; its deficit is $(a-1)/4 \leq m/8$ because $2(a-1) \leq m$. For $\ell = 2$, $\mathrm{OPT} = q = m - 1 = 2a + b + s$ and its deficit relative to q is $(a-1)/4 \leq q/8$. Thus both unbalanced path types have ratio at least $7/8$.

For $\ell = 1$, $\mathrm{OPT} = m = 2a + b + s$. Its deficit is $(a+1)/4$, which is at most $m/4$ since $m \geq 2a \geq a + 1$. On a cycle, the deficit is at most $a/4 \leq m/8$, since $m = 2a + b$. Sum these bounds over final components. Interleaving does not change their induced orders or their contributions. □

Lemma 5. *For every allowed advice table,* T *produces at least* $\mathrm{OPT}/2$ *matches on* $\mathcal{D}$ *and at least* $3\,\mathrm{OPT}/4$ *matches on* C*, deterministically.*

Proof. The output is maximal: an edge between two finally unmatched vertices would have been available when its online endpoint arrived. Every optimum edge touches an output edge, and each output edge can account for at most two optimum edges. This proves the $1/2$ bound.

On a cycle the boundary accounting holds deterministically for T. Each of its a roots creates exactly one occupied port, regardless of advice, and extensions preserve ports. Each failed merge consumes two occupied ports that cannot be used again. Hence there are at most $\lfloor a/2 \rfloor$ failures and

$$|\mathsf{T}(\text{cycle})| \geq m - \lfloor a/2 \rfloor \geq m - a/2 \geq 3m/4,$$

using $m = 2a + b$. Since OPT $= m$ for each cycle, summing proves the claim. The bound permits arbitrary correlations between root choices induced by the advice table. □

4 Interpolation and matching upper bounds

The two algorithms give the endpoints in Table 1. The advantage of cycles persists when advice is arbitrarily wrong, rather than only when the table is close to a maximum matching.

Table 1. Endpoints used to attain the exact frontiers. Pairs are (consistency, robustness); all bounds are multiplicative.

Realized class	Advice-free E	Advice rule T	Upper bounds
Maximum degree two	$(3/4, 3/4)$	$(1, 1/2)$	$r \leq 3/4$, $c + r \leq 3/2$
2-regular	$(7/8, 7/8)$	$(1, 3/4)$	$r \leq 7/8$, $c + r \leq 7/4$

Achievability. Write $\beta = 3/4$ on $\mathcal{D}$ and $\beta = 7/8$ on $\mathcal{C}$. Corollary 4 gives (β, β), while Lemmas 2 and 5 give $(1, 2\beta - 1)$. For $c \leq \beta$, use E. For $c \geq \beta$, choose T once before processing the input with probability

$$p = \frac{c - \beta}{1 - \beta},$$

and use E otherwise. For every fixed input and table, the mixture has consistency at least $(1 - p)\beta + p = c$ and robustness at least $(1 - p)\beta + p(2\beta - 1) = 2\beta - c$.

Path upper bounds. For completeness, specialize the known Z-path obstruction of [CGLB24, JM24] to the matching-table domain. Take $U = \{0, 1\}$, online order x, y, and $N(x) = \{0, 1\}$ in two worlds. Complete with $N_0(y) = \{0\}$ and $N_1(y) = \{1\}$. Both graphs belong to $\mathcal{D}$ and have optimum two. With a common advice table, the first action is the same when the algorithm's random tape is coupled across worlds. If it matches x, the sum of the best final cardinalities is three; if it abstains, the sum is two. Thus the sum of expected outputs is at most three.

Use empty advice in both worlds. Robustness implies $4r \leq 3$. Alternatively use the same table $A(x) = 0, A(y) = 1$ in both. This table is optimal in world 1 and allowed in world 0, so $2c + 2r \leq 3$.

Cycle upper bounds with a common table. Use the eight-cycle pair of Fàbrega and Muñoz [FM18], Proposition 1. Take $U = \{0, 1, 2, 3\}$ and online order x_1, x_2, y_1, y_2. The first two neighborhoods are $N(x_1) = \{0, 1\}$ and $N(x_2) = \{2, 3\}$ in both worlds. The remaining neighborhoods are

	$N(y_1)$	$N(y_2)$
G_0	$\{0, 2\}$	$\{1, 3\}$
G_1	$\{0, 3\}$	$\{1, 2\}$.

Each graph is a connected simple 2-regular bipartite graph with optimum four. Fix a common advice table and couple the random tape. If the two early vertices both match, one free offline vertex remains in each early pair. In exactly one world those two free vertices belong to the same late neighborhood, allowing at most one more match; in the other world at most two more matches are possible. The sum of final cardinalities is at most seven. With one or zero early matches the sums are at most six or four. This argument allows abstention, correlated choices, and arbitrary later strategies. Consequently the sum of expected outputs is at most seven.

Empty common advice gives $8r \le 7$. For the consistency bound, use

$$(A(x_1), A(x_2), A(y_1), A(y_2)) = (0, 2, 3, 1).$$

It is a perfect matching of G_1 and an allowed injective table for G_0. Only the pair $(y_1, 3)$ is absent in G_0; $(y_2, 1)$ is present in both worlds. Applying consistency in G_1 and robustness in G_0 gives $4c + 4r \le 7$. These fixed input pairs prove both upper bounds in Theorem 1; the preceding mixture attains them.

5 Checks and further questions

The accompanying standard-library checker directly enumerates root coin strings, maximum matchings, and injective advice tables. It verifies both component expectation formulas on paths and cycles through six online vertices, and checks all labelled equal-part graphs of size three under both degree bounds and all online orders. On these inputs it tests every maximum matching as advice; on regular inputs it also tests every allowed table, including absent-edge advice. It checks the upper gadgets by exact optimal continuation. These finite checks supplement the analytic proofs.

The formulas resolve the matching-table tradeoff under the stated oblivious input model. Further questions include adaptive adversaries, weighted matching, and a larger offline degree bound. The boundary-bit invariant and the decreasing-arrival proof both use the second appearance as the last possible appearance, so extensions to those models need additional ideas.

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